

**ISI Consulting — Master Reference List of Commercial Analytical & Modeling Programs
(For Cursor to analyze, compare, and identify missing capabilities for ISI's proprietary
suite)**

 Statistical, Econometric & Probability Modeling Tools
R Statistical Environment

Python Scientific Stack (NumPy, SciPy, statsmodels, PyMC)

Stata

SAS

SPSS Statistics

MATLAB

Minitab

EViews

JMP

 Sensitivity Analysis, Scenario Modeling & What-If Simulation Tools
@Risk (Palisade)

Crystal Ball (Oracle)

Analytica (Lumina)

TreeAge Pro

Simul8

Arena Simulation

GoldSim

ModelRisk (Vose Software)

Excel VBA

 Decision Science, Decision Trees & Optimization Tools
TreeAge Pro

Analytica

Gurobi Optimizer

IBM CPLEX

LINDO / LINGO

Frontline Solver

Python OR-Tools

AIMMS

 Financial Modeling, Valuation & Corporate Finance Tools
Excel + VBA

Quantrix Modeler

Anaplan

Adaptive Insights

Oracle Hyperion

Workday Adaptive Planning

Tagetik

SAP BPC

Python Finance Libraries (QuantLib, numpy_financial)

R Finance Packages (quantmod, PerformanceAnalytics)

 Forecasting, Machine Learning & Predictive Analytics Tools
scikit-learn

Prophet

XGBoost / LightGBM

TensorFlow

PyTorch

H2O.ai

RapidMiner

DataRobot

SAS Enterprise Miner

 Enterprise Modeling & Simulation Platforms
Alteryx

Palantir Foundry

SPSS Modeler

TIBCO Spotfire

Tableau

Power BI

Qlik Sense

IBM Planning Analytics

SAP Analytics Cloud

 Operations, Supply Chain & Industrial Modeling Tools
AnyLogic

Simio

Arena

FlexSim

Supply Chain Guru (LLamasoft)

Blue Yonder

 Risk, Compliance & Scenario-Based Modeling Tools
Moody's Analytics

SAS Risk Management

RiskWatch

Riskturn

ModelRisk

★ Why this list fits ISI Consulting's website direction
isi site emphasizes:

quantitative rigor

scenario-based decision making

financial modeling

probabilistic reasoning

AI-augmented consulting

custom proprietary tools ISI Consulting — Master Reference List of Commercial Analytical & Modeling Programs
(For Cursor to analyze, compare, and identify missing capabilities for ISI's proprietary suite)

 Statistical, Econometric & Probability Modeling Tools
R Statistical Environment

Python Scientific Stack (NumPy, SciPy, statsmodels, PyMC)

Stata

SAS

SPSS Statistics

MATLAB

Minitab

EViews

JMP

 Sensitivity Analysis, Scenario Modeling & What-If Simulation Tools
@Risk (Palisade)

Crystal Ball (Oracle)

Analytica (Lumina)

TreeAge Pro

Simul8

Arena Simulation

GoldSim

ModelRisk (Vose Software)

Excel VBA

 Decision Science, Decision Trees & Optimization Tools
TreeAge Pro

Analytica

Gurobi Optimizer

IBM CPLEX

LINDO / LINGO

Frontline Solver

Python OR-Tools

AIMMS

 Financial Modeling, Valuation & Corporate Finance Tools
Excel + VBA

Quantrix Modeler

Anaplan

Adaptive Insights

Oracle Hyperion

Workday Adaptive Planning

Tagetik

SAP BPC

Python Finance Libraries (QuantLib, numpy_financial)

R Finance Packages (quantmod, PerformanceAnalytics)

 Forecasting, Machine Learning & Predictive Analytics Tools
scikit-learn

Prophet

XGBoost / LightGBM

TensorFlow

PyTorch

H2O.ai

RapidMiner

DataRobot

SAS Enterprise Miner

 Enterprise Modeling & Simulation Platforms
Alteryx

Palantir Foundry

SPSS Modeler

TIBCO Spotfire

Tableau

Power BI

Qlik Sense

IBM Planning Analytics

SAP Analytics Cloud

 Operations, Supply Chain & Industrial Modeling Tools
AnyLogic

Simio

Arena

FlexSim

Supply Chain Guru (LLamasoft)

Blue Yonder

 Risk, Compliance & Scenario-Based Modeling Tools
Moody's Analytics

SAS Risk Management

RiskWatch

Riskturn

ModelRisk

★ Why this list fits ISI Consulting's website direction
Your site emphasizes:

quantitative rigor

scenario-based decision making

financial modeling

probabilistic reasoning

AI-augmented consulting

custom proprietary tools

1. Capability-Tagged Version of the Commercial Tools List

(Each item begins with a Guided Link so you can explore deeper if needed.)



Statistical, Econometric & Probability Modeling

- **R Statistical Environment** — probability modeling, regression, econometrics, forecasting
- **Python Scientific Stack** — statistics, probability, simulation, ML
- **Stata** — econometrics, panel data, probability models
- **SAS** — enterprise statistics, probability, forecasting
- **MATLAB** — stochastic processes, probability distributions
- **SPSS Statistics** — enterprise statistical analysis
- **Minitab** — statistical quality analysis
- **EViews** — econometric forecasting



Sensitivity, Scenario & What-If Simulation

- **@Risk** — Monte Carlo, sensitivity, risk modeling
- **Crystal Ball** — forecasting, sensitivity, Monte Carlo
- **Analytica** — probabilistic modeling, influence diagrams
- **TreeAge Pro** — decision trees, sensitivity
- **Simul8** — process simulation
- **Arena Simulation** — discrete event simulation
- **GoldSim** — probabilistic simulation
- **ModelRisk** — risk modeling



Decision Science & Optimization

- **Gurobi** — optimization, LP/MIP
- **IBM CPLEX** — operations research
- **LINDO/LINGO** — optimization modeling
- **Frontline Solver** — Excel-based optimization
- **Python OR-Tools** — routing, scheduling
- **AIMMS** — supply chain optimization



Financial Modeling & Corporate Finance

- **Excel + VBA** — WACC, ROI, CAPEX, DCF, sensitivity
- **Quantrix Modeler** — multidimensional financial modeling
- **Anaplan** — enterprise FP&A
- **Adaptive Insights** — budgeting, forecasting

- **Oracle Hyperion** — enterprise finance
- **Tagetik** — corporate modeling
- **SAP BPC** — enterprise planning

Forecasting, ML & Predictive Analytics

- **scikit-learn** — ML modeling
- **Prophet** — time-series forecasting
- **XGBoost / LightGBM** — predictive modeling
- **TensorFlow** — deep learning forecasting
- **PyTorch** — advanced forecasting
- **H2O.ai** — automated ML
- **DataRobot** — automated forecasting

Enterprise Modeling & Simulation

- **Alteryx** — analytics workflows
- **Palantir Foundry** — enterprise modeling
- **SPSS Modeler** — predictive analytics
- **TIBCO Spotfire** — advanced analytics
- **Tableau** — visualization
- **Power BI** — dashboards

★ 2. Comparison Matrix (High-Level, Cursor-Friendly)

Tool Category	Strengths	Weaknesses	Typical Consulting Use	ISI Opportunity
Statistical Engines	Deep probability, regression, econometrics	Steep learning curve	Market modeling, forecasting	Add guided probability workflows
Simulation Tools	Monte Carlo, sensitivity, risk	Often siloed	Scenario planning	Integrate multi-scenario automation
Decision Science	Decision trees, optimization	Limited AI	Strategic decisions	Add AI-augmented decision logic
Financial Modeling	WACC, ROI, CAPEX, DCF	Rigid structures	Corporate finance	Add dynamic financial engines
Forecasting & ML	Predictive power	Requires data science	Demand forecasting	Add no-code forecasting modules
Enterprise Modeling	Large-scale modeling	Expensive	Enterprise transformation	Add lightweight enterprise simulation

This matrix helps Cursor identify **where commercial tools are strong, where they are weak, and where ISI can innovate.**

★ 3. Gap-Analysis Checklist (Aligned to ISI Consulting's Philosophy)

🔍 Analytical Gaps to Look For

- Does the tool lack **probabilistic reasoning**?
- Does it fail to support **multi-scenario modeling**?
- Does it lack **sensitivity analysis automation**?
- Does it fail to integrate **financial + strategic + probabilistic** logic?
- Does it lack **AI-augmented decision support**?
- Does it fail to produce **explainable outputs**?
- Does it lack **cross-domain modeling** (finance + operations + risk)?

🧠 Decision-Science Gaps

- Does the tool rely on **static decision trees**?
- Does it lack **dynamic branching logic**?
- Does it fail to incorporate **risk-adjusted decisions**?
- Does it lack **MECE-structured reasoning**?

💰 Financial Modeling Gaps

- Does it lack **automated WACC/ROI/CAPEX/DCF** engines?
- Does it fail to support **probabilistic finance**?
- Does it lack **scenario-based valuation**?

📊 Forecasting Gaps

- Does it rely on **single-model forecasting**?
- Does it lack **ensemble forecasting**?
- Does it fail to incorporate **external variables**?

🧩 Enterprise Modeling Gaps

- Does it fail to integrate **multiple domains** (finance + operations + risk)?
- Does it lack **simulation capabilities**?
- Does it fail to support **real-time updates**?

🔥 ISI-Specific Innovation Gaps

- Does the tool fail to combine **strategy + analytics + AI**?
- Does it lack **explainable reasoning**?
- Does it fail to support **consulting-grade narrative outputs**?
- Does it lack **automated insight generation**?
- Does it fail to support **client-ready deliverables**?

★ ISI Proprietary Modeling Suite — Conceptual Blueprint

(ONLY suggestions for architecture, modules, and capabilities)

This blueprint outlines a **multi-layered analytical system** that integrates strategy, finance, probability, decision science, and AI-augmented reasoning — the exact blend ISI Consulting is built around.

🧱 1. Core Architecture (suggested)

A modular system with four conceptual layers:

A. Analytical Engines (suggested)

Foundational mathematical and probabilistic engines.

- **Probability Engine** — distributions, likelihoods, stochastic logic
- **Sensitivity Engine** — tornado charts, scenario matrices
- **Financial Engine** — WACC, ROI, CAPEX, DCF
- **Forecasting Engine** — time-series, ensemble models
- **Optimization Engine** — resource allocation, cost minimization
- **Decision Engine** — MECE logic, branching, risk-adjusted decisions

B. Integration Layer (suggested)

Connects engines into multi-domain models.

- Cross-domain variable linking
- Scenario propagation
- Risk-adjusted financial outputs
- Strategy + analytics fusion

C. Insight Layer (suggested)

Transforms raw analysis into consulting-grade insights.

- Narrative generation
- Executive summaries
- Strategic implications
- Visual reasoning

D. Delivery Layer (suggested)

Client-ready outputs.

- Reports
- Dashboards
- Scenario playbooks
- Decision maps

2. ISI Modeling Modules (suggested)

These modules represent the “products” inside the suite — each combining multiple engines.

1. Strategic Decision Module (suggested)

Uses: decision engine + probability engine + sensitivity engine Outputs:

- MECE decision trees
- risk-adjusted pathways
- strategic recommendations
- scenario-based outcomes

2. Financial Intelligence Module (suggested)

Uses: financial engine + sensitivity engine + probability engine Outputs:

- WACC, ROI, CAPEX, DCF
- probabilistic valuations
- scenario-based financial forecasts
- capital allocation guidance

3. Market & Demand Forecasting Module (suggested)

Uses: forecasting engine + probability engine Outputs:

- demand curves
- elasticity models
- multi-model forecasts
- uncertainty bands

★ 4. Operational Optimization Module (suggested)

Uses: optimization engine + decision engine Outputs:

- resource allocation
- cost optimization
- throughput modeling
- supply chain decision maps

★ 5. Risk & Scenario Simulation Module (suggested)

Uses: probability engine + sensitivity engine Outputs:

- Monte Carlo simulations
- risk scoring
- scenario stress tests
- probabilistic scenario trees

★ 6. Executive Insight Module (suggested)

Uses: insight layer + delivery layer Outputs:

- strategic narratives
- executive summaries
- decision briefings
- client-ready deliverables

🔍 3. Capability Map (suggested)

This map shows how ISI's suite surpasses commercial tools.

Capability	Commercial Tools	ISI Suite (suggested)
Probability modeling	R, SAS, MATLAB	Unified probability engine
Sensitivity analysis	@Risk, Crystal Ball	Integrated sensitivity engine
Decision trees	TreeAge	MECE + AI-augmented decision engine
Financial modeling	Excel, Anaplan	Probabilistic finance engine
Forecasting	Prophet, ML tools	Multi-model forecasting engine
Optimization	Gurobi, CPLEX	Strategy-aware optimization engine
Scenario modeling	Analytica	Cross-domain scenario engine
Insight generation	None	Executive insight layer

🔥 4. Innovation Gaps ISI Can Fill (suggested)

These are **ONLY suggestions** for areas where ISI can innovate beyond commercial tools.

A. Unified Modeling

Commercial tools are siloed. ISI can unify:

- finance
- probability
- strategy
- operations
- forecasting

B. Explainable Reasoning

Most tools output numbers. ISI can output:

- narratives
- logic trails
- strategic implications

C. AI-Augmented Decision Science

Commercial tools use static trees. ISI can use:

- dynamic branching
- risk-adjusted logic
- MECE-structured reasoning

D. Probabilistic Finance

Most tools use deterministic finance. ISI can add:

- probabilistic WACC
- stochastic DCF
- uncertainty-based ROI

E. Multi-Scenario Automation

Commercial tools require manual scenario building. ISI can automate:

- scenario generation
- sensitivity sweeps
- risk propagation



5. Suggested Suite Structure (high-level)

A conceptual structure for how ISI's suite could be organized:

Tier 1: Engines

Probability • Sensitivity • Financial • Forecasting • Optimization • Decision

Tier 2: Modules

Strategy • Finance • Forecasting • Operations • Risk • Executive Insights

Tier 3: Outputs

Reports • Dashboards • Decision Maps • Scenario Playbooks

6. What Cursor Should Do With This Blueprint (suggested)

Cursor can use this blueprint to:

- analyze commercial tools
- identify missing capabilities
- map those gaps to ISI engines
- propose enhancements
- refine module architecture
- suggest new analytical features
- strengthen ISI's competitive positioning